

1 Safety of Repeated Low-Level Red-Light Therapy for Myopia: A Systematic 2 Review

3

4 **Running Title:** Safety of repeated low-level red-light therapy

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6 Authors

7 Yanping Chen, MD^{1*}, Shida Chen, MD, PhD^{1*}, Ruilin Xiong, MD, PhD^{1*}, Shaopeng

8 Yang, MD¹, Riqian Liu, MD¹, Ziyu Zhu, MBBS¹, Kaidi Xiang, MD^{2,3}, Nathan

9 Congdon, MD, MPH^{1,4,5†}, Wei Wang, MD, PhD^{1†}

10

11 Affiliations

12 1. State Key Laboratory of Ophthalmology, Zhongshan Ophthalmic Center, Sun
13 Yat-sen University, Guangdong Provincial Key Laboratory of Ophthalmology and
14 Visual Science, Guangdong Provincial Clinical Research Center for Ocular
15 Diseases, Guangzhou, Guangdong, China.

16 2. Department of Clinical Research, Shanghai Eye Disease Prevention and
17 Treatment Center, Shanghai Eye Hospital, Shanghai Vision Health Center and
18 Shanghai Children Myopia Institute, Shanghai, China.

19 3. Department of Ophthalmology, Shanghai General Hospital (Shanghai First
20 People's Hospital), Shanghai Jiao Tong University School of Medicine, Shanghai,
21 China; National Clinical Research Center for Eye Diseases; Shanghai Clinical
22 Research Center for Eye Diseases; Shanghai Key Clinical Specialty; Shanghai
23 Key Laboratory of Ocular Fundus Diseases; Shanghai Engineering Center for
24 Visual Science and Photomedicine; Shanghai Engineering Center for Precise
25 Diagnosis and Treatment of Eye Diseases, Shanghai Eye Research Institute,
26 Shanghai, China.

27 4. Centre for Public Health, Queen's University Belfast, Belfast, United Kingdom.

28 5. Orbis International, New York, New York, United States of America.

29 *Co-first authors. †Co-corresponding authors.

30

31 **Corresponding Author**

32 Wei Wang, MD PhD, Zhongshan Ophthalmic Center, State Key Laboratory of

33 Ophthalmology, Sun Yat-sen University, Guangzhou, China.

34 Email: wangwei@gzzoc.com | ORCID: <https://orcid.org/0000-0002-5273-3332>

35

36 Nathan Congdon, MD, MPH, Zhongshan Ophthalmic Center, State Key Laboratory of

37 Ophthalmology, Sun Yat-sen University, Guangzhou, China; Centre for Public Health,

38 Queen's University Belfast, Belfast, United Kingdom; Orbis International, New York,

39 New York, United States of America.

40 Email: ncongdon1@gmail.com | ORCID: <https://orcid.org/0000-0001-9866-3416>

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49

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55 **Abbreviations and Acronyms:**

56 RLRL = repeated low-level red-light; OCT = optical coherence tomography; CI =

57 confidence interval; IMI = International Myopia Institute; RCT = randomized

58 controlled trial; SER = spherical equivalent refraction; AL = axial length; NNH =

59 number needed to harm; NNT = number needed to treat; SVS = single-vision

60 spectacles; BCVA = best-corrected visual acuity.

62 Abstract

63 **Topic:** Existing evidence for the safety of repeated low-level red-light (RLRL)
64 therapy for myopia control.

65 **Clinical relevance:** Recent trials show RLRL therapy is effective in the prevention
66 and control of myopia. Establishing its safety profile is necessary prior to widespread
67 clinical implementation.

68 **Methods:** We conducted a systematic review (International Prospective Register of
69 Systematic Reviews, CRD42024516676) of articles across seven databases from
70 inception through February 10, 2024, with keywords related to myopia and RLRL
71 therapy. Pooled safety outcomes and risk-to-benefit ratios were reported, and
72 incidence of side effects was compared with other anti-myopia interventions. Quality
73 appraisal was performed using the Cochrane Risk of Bias Tool.

74 **Results:** Among 689 screened articles, 20 studies (2.90%; eleven randomized
75 controlled trials, four non-randomized controlled trials, one post-trial study, one
76 single-arm study, one retrospective study and two case reports of identical patient.;
77 median duration 9 months, longest 24 months) were analysed, encompassing 2,380
78 participants aged 3-18 years and 1,436 individuals undergoing RLRL therapy. Two
79 case reports described an identical patient with reversible decline in visual acuity and
80 optical coherence tomography (OCT) abnormalities, completely resolved 4 months
81 after treatment cessation. No cases of permanent vision loss were reported. Temporary
82 afterimage was the most common ocular symptom following treatment, resolving
83 within 6 minutes in reported studies. The number needed to harm outweighed the
84 number needed to treat by a ratio of 12.7-21.4 for a person with -3D to -8D myopia
85 treated with RLRL therapy. Incidence of side effects from RLRL was 0.088 per 100
86 patient-years (95% confidence interval [CI], 0.02-0.50), comparable to spectacles
87 designed for myopia reduction (0.22; 95% CI, 0.09-0.51; P=0.385), and significantly
88 lower than for low-dose atropine (7.32; 95% CI, 6.65-8.05; P<0.001), orthokeratology
89 (20.6; 95% CI, 16.7-25.0; P<0.001), other anti-myopia contact lens (19.3; 95% CI,
90 17.6-21.1; P<0.001).

91 **Conclusion:** No irreversible visual function loss or ocular structural damage was

92 identified with RLRL. Fundus photography and OCT before and during therapy,
 93 alongside home monitoring of visual acuity and duration of afterimages, are necessary
 94 to identify side effects. Further adequately-powered studies of longer duration are
 95 needed to evaluate long-term safety of RLRL.

96

97 **Keywords:** Refractive error, myopia prevention, repeated low-level red-light, safety,
 98 risk-to-benefit

99

100 **Introduction**

101 Myopia is the most common ocular disorder of childhood and adolescence. Without
 102 effective intervention measures, it is estimated that approximately 50% of the global
 103 population will be affected by 2050.¹ The rising prevalence of myopia, along with
 104 earlier onset, increases the risk of high myopia, which may be associated with
 105 irreversible, sight-threatening complications such as myopic maculopathy, glaucoma
 106 and retinal detachment.²⁻⁴ Therefore, prevention and control of myopia have become
 107 important public health challenges.

108

109 Repeated low-level red-light (RLRL) therapy, highlighted in the latest 2023 Digest of
 110 the International Myopia Institute (IMI), has emerged as a novel approach to myopia
 111 prevention.⁵ RLRL therapy involves locally irradiating the retina with low-level red
 112 light (approximately 1600 lux). It falls under Group 1 of the ANSI Z80.36-2021
 113 standard,⁶ ensuring its safety for clinical ophthalmic applications. Unlike traditional
 114 therapeutic lasers such as panretinal photocoagulation, which rely on thermal effects
 115 generated through transpupillary energy ranging from 200-250 mW passing through
 116 the pupil, RLRL uses a much lower energy level of 0.29 mW, avoiding any thermal
 117 effects. Its therapeutic impact is postulated to rely on photobiomodulation (PBM)
 118 secondary to the laser's energy, potentially leading to thickening of the choroidal
 119 layer.⁷

120

121 Growing trial evidence supports the effectiveness of RLRL therapy in reducing
 122 myopia progression. The first multicenter randomized controlled trial (RCT) in China
 123 demonstrated in 2019 that home-based RLRL therapy, administered for 3 minutes
 124 twice daily for 5 days a week, significantly reduced axial elongation by 69.4% and
 125 refractive progression by 76.6% among school-aged children.⁸ Subsequent clinical
 126 trials have consistently confirmed these findings.⁹⁻¹¹ Recently, a meta-analysis
 127 incorporating 13 studies comprising 8 RCTs, 3 non-randomized controlled trials and 2
 128 cohort studies, and involving a total of 1857 children and adolescents, confirmed the
 129 efficacy of RLRL therapy.¹² Six-month weighted differences in spherical equivalent

refraction (SER) and axial length (AL) were 0.68 diopters (D) and 0.35 mm respectively between the RLRL treatment group and controls. However, published studies have predominantly focused on efficacy outcomes, with limited reporting of the safety and side effects of RLRL therapy.

To address this important knowledge gap, we systematically reviewed the safety profiles and risk-to-benefit ratio associated with RLRL treatment for myopia prevention and control. The aim of this study is to provide insights to help both clinicians and program designers optimize use of RLRL therapy, ensuring its safe and effective integration into myopia management.

Methods

This review was registered prospectively on the International Prospective Register of Systematic Reviews database (available from https://www.crd.york.ac.uk/prospero/display_record.php?ID=CRD42024516676) and is reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement.¹³ All research adhered to the tenets of the Declaration of Helsinki. Individual patient-level consent was not required, nor was ethical review.

Eligibility Criteria

We included clinical studies of RLRL therapy designed to prevent myopia or delay its progression. Studies were selected according to the following criteria: (1) Participants were younger than 18 years with myopia or premyopia; (2) RLRL therapy was used in at least one arm or by all participants; (3) Reporting of at least 1 safety outcome, including visual function, ocular structural assessment, or adverse events; (4) All study types except literature reviews, in order to capture as many reported safety profiles and side effects as possible. We excluded studies on non-human subjects, those enrolling participants with secondary myopia, without safety data, and those merely describing treatments combined with RLRL therapy or involving red light

160 flicker rather than continuous administration.

161

162 **Search Strategy**

163 A comprehensive search of the peer-reviewed literature was conducted across seven
164 databases, including PubMed, Embase, Web of Science, Cochrane Library, Scopus,
165 the Chinese databases China National Knowledge Infrastructure and VIP Information
166 Database, from their dates of inception through 10 February 2024. We used a
167 combination of key words related to myopia and RLRL therapy, imposing no
168 language restrictions. To ensure thorough coverage, we also scrutinized relevant
169 reviews and all references cited by eligible studies for additional pertinent
170 publications. The full search strategy is detailed in **Table S1**.

171

172 **Study Selection**

173 Citations retrieved from electronic databases were compiled into an EndNote library
174 by one author (Y.C.). After the removal of duplicates, two reviewers (Y.C., R.X.)
175 independently screened titles and abstracts to assess initial eligibility. Reviewers then
176 checked the full text for potentially eligible studies to determine their final inclusion
177 or exclusion. The primary reason for exclusion was documented at the full text
178 screening stage. Any discrepancies between reviewers were resolved through
179 discussion or by consulting a third researcher if necessary (W.W.).

180

181 **Data Collection and Risk of Bias Assessment**

182 Data were extracted independently by two reviewers (Y.C., R.X.)
183 and were entered into an Excel spreadsheet (version 2022, Microsoft Corporation,
184 Redmond, USA). For each included study, the extracted information consisted of
185 author's name, year of publication, study design, country or area, specification of the
186 red light device (device name, manufacturer, wavelength and power), treatment
187 regimen, sample size, follow-up duration, age, sex, baseline SER, baseline AL, safety
188 outcomes, and participation completion rate. For the safety outcomes, we documented
189 all safety data reported throughout the selected studies. Two reviewers (Y.C., S.Y.)

independently appraised articles for systematic bias using the Cochrane Risk of Bias Tool.¹⁴

Data Synthesis and Analysis

Given the heterogeneous reporting of safety outcomes among the included studies, it was not feasible to conduct a pooled meta-analysis.¹⁵ Instead, we performed a risk-to-benefit analysis according to the model described by Bullimore to assess whether the potential benefits of reducing myopia progression of 1D with RLRL therapy outweigh the potential risks associated with the treatment.¹⁶ The revised Bullimore's model is based on three assumptions: 1) Every patient undergoing myopia control will use RLRL for a 5-year treatment period. Since there's lack of long-term efficacy of RLRL therapy, the selection of five years facilitates the conservative estimation of the effect of controlling myopia progression within 1D as per previous review.¹⁷ 2) Serious adverse events, if they occur, may happen at a mean age of 12 years during this five-year treatment period.¹⁶ 3) Based on the estimated mean life expectancy of 77 years in China (<https://data.who.int>), any adverse event resulting in immediate visual impairment will lead to visual impairment for a duration of 65 years. The annual incidence rate of vision loss was calculated using the number of children experiencing visual impairment from RLRL as the numerator and the estimated total number of patient-years in reported clinical RLRL studies with safety outcomes as the denominator. The absolute risk increase (ARI) of vision loss years was estimated by multiplying the annual incidence of vision loss per 10,000 patients by the assumed duration of visual impairment, which is 65 years. The number needed to harm (NNH) was then calculated as the reciprocal of ARI. This metric indicates the number of patients who need to be treated to induce one case of visual impairment.

According to Bullimore's model, the benefit of preventing visual impairment from blinding myopic related complications by 1D was regardless of the treatment. The average duration of visual impairment that a patient is likely to experience over their lifetime at myopia ranging from -3D to -8D was estimated.¹⁶ For instance, a patient

with -3D myopia is expected to experience an average of 4.42 years of visual impairment (mild visual impairment as US definition of 20/40) while a -4D person will experience 5.25 years of visual impairment. Thus, the benefit of slowing myopia progression by 1D is quantified by the difference in years of visual impairment, amounting to a prevention of 0.84 years of visual impairment if myopia control interventions in an individual potentially reaching -4D myopia result in achieving only -3D. The values of years of visual impairment prevented by 1D reduction ranges from 0.74 to 1.22 for -3D to -8D. The number needed to treat (NNT) was evaluated in Bullimore's model to prevent 1-year (NNT range, 0.82-1.38) and 5-year (NNT range, 4.11-6.75) visual impairment.¹⁶ Furtherly, we calculated the NNH/NNT ratio for RLRL therapy in patients with myopia degrees ranging from -3D to -8D.

Additionally, we conducted a systematic comparison of the incidence of side effects between RLRL therapy and other anti-myopia interventions. Our approach involved a systematic search of all eligible peer-reviewed RCTs that included myopic or premyopic participants younger than 18 years, had at least a one-year follow-up period, and investigated interventions including low-dose atropine, orthokeratology, other contact lenses, and spectacles. For spectacles, we included bifocal lens, progressive addition spectacles, aspherical lenslets and peripheral defocus spectacles as spectacles designed for myopia reduction since their incidence rates of adverse events were similar. Adverse events of other interventions were counted as the number of events reported in published RCTs. The crude incidence of adverse events was computed per 100 patient-years of intervention, with 95% confidence intervals (CIs) calculated using the Wilson method.^{18,19} A two-sided p value <0.05 was defined as statistically significant. All data analyses were performed using Stata (version 17.0, StataCorp, College Station, Texas, USA).

Results

Study Selection

Of the 689 references retrieved, 39 full-text articles (5.66%) were reviewed, and 20

studies (2.90%) were identified as eligible for systematic review (**Figure 1**). The primary reason for exclusion during the full-text review process was the absence of reported safety outcomes (n=14, 73.7%).^{7, 20-32}

Characteristics of Included Studies

The 20 studies comprised eleven RCTs,^{8, 10, 11, 33-40} four non-randomized controlled trials,⁴¹⁻⁴⁴ one post-trial study,⁹ one single-arm study,⁴⁵ one retrospective study⁴⁶ and two case reports of identical patient.^{47, 48} Fifteen studies evaluated the safety and efficacy of RLRL therapy compared to controls, eleven with single-vision spectacles (SVS),^{8, 10, 33, 35-37, 39-43} one with a sham device,¹¹ one with 0.01% atropine,³⁴ and one with orthokeratology, SVS, and combination treatment of orthokeratology and RLRL.⁴⁴ One study compared the safety and efficacy between RLRL devices with different powers and SVS.³⁸ Two case reports described the identical individual case with different details;^{47, 48} thus we comprehensively reviewed the two reports but only included as one participant in the systematic review. These studies included a total of 2,380 participants aged 3-18 years presenting at baseline with myopia (cycloplegic SER -0.50 to -9.00D) or premyopia (cycloplegic SER -0.50 to +0.75D), among them 1,436 subjects undergoing RLRL therapy. The median follow-up duration was 9 months, with an interquartile range of 6-12 months, and a longest follow-up period of 24 months. All studies reported a participant completion rate of over 50%, were conducted in China, and published between 2021 and 2024. (**Table 1**). The risk of bias assessment is shown in **Figure 2**.

The specifications of RLRL therapy and treatment regimens across the included studies are summarized in **Table S2**. Four types of RLRL devices were utilized, including Eyerising (Jiangsu, China; n=14 studies; 650±10 nm, 2.00±0.50 mW),^{8-11, 33, 34, 40-42, 44-48} LD-A (Jilin, China; n=2 studies; 635 nm, 0.35±0.02 mW),^{35, 37} YF020A (Hunan, China; n=3 studies; 650 nm, no specified power),^{36, 39, 43} and sky-n1201 (Beijing, China; n=1 study; 650 nm, with variable outputs of 0.37±0.02 mW, 0.60±0.2 mW, and 1.20 mW).³⁸ All studies adhered to a uniform daily RLRL treatment protocol,

consisting of two daily 3-minute sessions, with a minimum interval of 4 hours
between sessions. Six studies implemented a 5-day per week regimen,^{8-10, 41, 43, 46}
while the remaining 14 studies provided treatment 7 days per week.^{11, 33-40, 42, 44, 45, 47, 48}

283

284 **Safety Profiles of RLRL Therapy**

285 **Table 2** presents the safety outcomes reported among participants undergoing RLRL
286 therapy. In these studies, visual function was assessed using best-corrected visual
287 acuity (BCVA) or multifocal electroretinogram, while ocular structures were
288 evaluated with optical coherence tomography (OCT), fundus photography, or
289 slit-lamp microscopy. Among 529 children assessed for visual function, 528 (99.8%)
290 either maintained a BCVA of $\geq 20/25$ or experienced no decline at follow-up visits
291 compared to baseline. OCT identified no structural damage in 624 (99.8%) of the 625
292 participants with such data.

293

294 Two case reports described the same twelve-year-old girl who experienced a two-line
295 decline in binocular BCVA after five months of RLRL therapy.^{47, 48} The child was
296 highly myopic prior to initiating RLRL, with SER of -6.75D in the right eye and
297 -6.25D in the left eye, and BCVA of 20/25 at the point of initiation. Due to repeated
298 inflammation from orthokeratology, the subject had been switched to RLRL for
299 managing myopia. After three months of RLRL treatment, the BCVA improved to
300 20/20. After five months for RLRL treatment, the child showed the binocular BCVA
301 decrease from 20/20 to 20/30 with prolonged afterimages occasionally exceeding 8
302 minutes. Multifocal electroretinogram indicated moderately and mildly decreased
303 response in the macula and paramacular respectively. OCT images showed bilateral
304 disruption of the foveal ellipsoid zone and discontinuity of the interdigitation zone.
305 Fundus photographs revealed bilaterally darkened foveae with a hypoautofluorescent
306 plaque in autofluorescence images. The bilateral outer retinal damage showed partial
307 recovery, with an improvement in binocular BCVA to 20/25, three months after
308 discontinuation of RLRL therapy. Complete recovery of bilateral outer retinal damage
309 was observed four months after discontinuing RLRL therapy.

310
 311 Nineteen studies reported on ocular adverse events. In one study with 20 participants
 312 treated with RLRL, afterimages lasting a mean of 3.2 ± 1.2 minutes were documented
 313 without any objective ocular abnormalities.³⁷ One trial reported afterimage as the
 314 most common phenomenon post light therapy, which can be alleviated by a short
 315 period of eye-closing and rest.⁴¹ Another study demonstrated no afterimages longer
 316 than 6 minutes after 1-year RLRL treatment within 126 premyopic children.¹⁰ One
 317 individual experienced dizziness following the red-light therapy; however, this
 318 symptom resolved within a few minutes and persisted during the immediate
 319 post-treatment period for only a few days.³⁴ Aside from the aforementioned case
 320 report,^{47, 48} no other included participants experienced vision loss of ≥ 2 lines, scotoma
 321 or treatment-related adverse events post-therapy.

322

323 **Risk-to-benefit Analysis**

324 The sole event of temporary vision loss described in the case reports is considered as
 325 the numerator,^{47, 48} and the denominator is a conservative estimate of the total number
 326 of participants in reported clinical RLRL studies with safety outcomes. The annual
 327 incidence of vision loss from RLRL therapy was estimated as 8.77 per 10,000
 328 patient-years. **Table 3** shows the NNH and NNH/NNT estimates for 1 year and 5
 329 years visual impairment associated with RLRL based on Bullimore's model. A total of
 330 17.5 individuals have to implement RLRL therapy for 5 years to result in 1-year of
 331 visual impairment, while 87.7 patients have to use it to result in 5-year of visual
 332 impairment.

333

334 From Bullimore's assessment, controlling myopia by 1D prevents between 0.74 and
 335 1.22 years of visual impairment due to myopic complications across myopia levels of
 336 -3D to -8D.¹⁶ This benefit outweighs the risk of visual impairment years from RLRL
 337 treatment, which is 570 per 10,000 patients, or 0.057 year per patient. On the other
 338 hand, the NNH outweighs the NNT by a ratio of 12.7-21.4 for a person with -3D to
 339 -8D myopia treated with RLRL therapy.

340

341 **Comparison with Other Interventions**

342 We further compared the incidence rates of ocular adverse events between RLRL and
343 other myopia interventions. The characteristics and adverse events of other treatments
344 reported in RCTs lasting at least 1 year are displayed in **Table S3**. The side effect
345 incidence of RLRL therapy is 0.088 per 100 patient-years (95% confidence interval
346 [CI], 0.02-0.50), which is comparable to spectacles designed for myopia reduction
347 (0.22 per 100 patient-years; 95% CI, 0.09-0.51; P=0.385),⁴⁹⁻⁵⁹ and significantly lower
348 than for low-dose atropine (7.32 per 100 patient-years; 95% CI, 6.65-8.05;
349 P<0.001),⁶⁰⁻⁷⁸ orthokeratology (20.6 per 100 patient-years; 95% CI, 16.7-25.0;
350 P<0.001)^{77, 79-84} and other anti-myopia contact lens (19.3 per 100 patient-years; 95%
351 CI, 17.6-21.1; P<0.001)⁸⁵⁻⁹⁰ (**Table 4**).

352

353 **Discussion**

354 The results of the systematic review indicated that no visual function loss or ocular
355 structural change with irreversible damage was identified with RLRL therapy. An
356 afterimage was the most common ocular symptom following treatment, with a
357 resolution time of less than 6 minutes reported in two clinical trials. This phenomenon,
358 induced by prior adaptation to a visual stimulus, is believed to be due to the natural
359 bleaching of photochemical pigments or neural adaptation in the retina. Participants
360 with an afterimage duration exceeding 6 minutes were considered clinically too
361 sensitive to the light stimulus.^{10, 91} Among dozens of published studies, only two
362 reported on a single case of a reversible decline in visual function and discontinuity of
363 the foveal ellipsoid and interdigitation zones in an identical girl, with complete
364 recovery after four months of treatment cessation. A risk-to-benefit analysis
365 emphasized that the benefit of reducing visual impairment outweighed the potential
366 risks associated with RLRL treatment. The incidence rate of side effects from RLRL
367 therapy was comparable to that of spectacles wearing, and was lower than that of
368 other therapies.

369

Of note, two case reports, both describing on complications of the same child following RLRL therapy offers valuable clinical data and serves as a reference point for RLRL's clinical application.^{47, 48} Given the rarity of this adverse event, it might be secondary to specific individual differences, where the patient may be especially responsive to light therapy. Such individual variability may render the retina more sensitive to light and prone to phototoxicity. It is important to note that the patient was highly myopic, a condition that often involves inherent issues with the retinal structure of the fundus. From the addition information provided by the authors of one case report,⁴⁸ this child experienced significant myopia regression (cycloplegic SER from -6.75D at baseline to -4.50D after 1 month in the right eye; from -6.25D at baseline to -4.50D after 1 month in the left eye) following RLRL treatment. The dark choroid shown in the OCT image post RLRL has been suggested in a non peer-reviewed letter to the editor regarding the case written by experts in inherited retinal disease to represent Stargardt's disease (<https://jamanetwork.com/journals/jamaophthalmology/fullarticle/2805391>).

Although the incidence of such adverse events is extremely low, identifying such super-responders to RLRL therapy is still important. From this case, possible characteristics of super-responders include a significant treatment effect, a marked SER regression or AL shortening and an afterimage duration exceeding 6 minutes in response to light exposure. It is crucial for clinicians and parents to closely monitor subjects with these characteristics, as any decrease in visual acuity while wearing glasses may suggest the development of such complications, which can then be further detected and characterized through OCT examination. Importantly, these complications are reversible as after stopping treatment for 4 months, the visual function and ocular structures changes returned to normal. Meticulous supervision is thus necessary throughout the treatment process to ensure safe implementation of RLRL. Appropriate actions include documentation of the retina through fundus photography and OCT before starting treatment and at each routine examination, tracking visual acuity, and recording the duration of any afterimages.⁹²

400

401 A comprehensive risk-to-benefit assessment of myopia treatments should consider

402 various factors, including the intervention effectiveness in slowing myopia

403 progression, the risk of myopia-related visual impairment, the degree of myopia

404 treated, and the specific risks associated with each intervention. From this

405 risk-to-benefit analysis, the risk of vision loss associated with RLRL treatment is seen

406 to be counterbalanced by its benefits in preventing myopia-related visual impairment

407 with a NNH/NNT ratio of 12.7 to 21.4. Previous studies have reported a NNH/NNT

408 ratio ranging from 5.43 for -3D to 9.15 for -8D for overnight contact lens wearing.^{16,}

409 ⁹³

410

411 Additionally, nearly all interventions for myopia control are associated with some

412 side effects and complications. Our systematic review summarized RCTs with at least

413 one year of follow-up reporting the incidence of side effects from available

414 anti-myopia interventions. Our findings indicate that the incidence of adverse events

415 associated with orthokeratology is 20.6 per 100 patient-years, while it was 19.3 per

416 100 patient-years for other contact lens. Complications linked to contact lens usage

417 included ocular noninfectious inflammatory events and sight-threatening microbial

418 keratitis.^{90, 94-96} A retrospective study authorized and approved by the US FDA

419 estimated the incidence of microbial keratitis from orthokeratology in children to be

420 14 per 10,000 patient-years (95% CI, 1.7-50.4 per 10,000 patient-years).⁹⁷ Atropine,

421 even at low concentrations of 0.01%, may lead to pupil dilation and loss of

422 accommodation, with photophobia, reduced near vision, and allergic conjunctivitis

423 being commonly reported ocular side effects.^{98, 99} Spectacles in comparison are a

424 more well-tolerated method for correcting and controlling myopia; however, they are

425 associated with a low risk of falls and bicycle collisions.^{51, 100} In this review, RLRL

426 therapy had a comparable incidence of side effects with spectacles.

427

428 This systematic review provides the first comprehensive evidence on the safety profile

429 and the risk-to-benefit ratio of RLRL therapy for myopia control. However, the results

430 should be interpreted within the context of several limitations. Firstly, most studies
 431 lasted for 12 months, with only 11 children undergoing 24-month treatment in one
 432 post-trial study included thereby limiting long-term evidence. There is a need for
 433 further large-scale studies to thoroughly assess the long-term safety of RLRL therapy
 434 in children and adolescents. Secondly, visual function in the included studies was
 435 primarily evaluated using visual acuity, which could be influenced by subjective
 436 factors. Objective assessments, such as multifocal electroretinography or
 437 microperimetry, are necessary to provide more comprehensive safety evidence for
 438 RLRL therapy. Thirdly, the methodology for reporting adverse events varied widely
 439 across studies. Some studies documented the number of patients experiencing side
 440 effects, whereas others detailed the number of adverse events. Fourthly, the accuracy
 441 of Bullimore's model is dependent on the validity of its assumptions.¹⁶ It presumes
 442 that the risk of vision loss from RLRL therapy is independent of refractive error. The
 443 model also assumes a fixed treatment effect with myopia control. Fifthly, quality
 444 appraisal was employed exclusively for the included RCTs, as the Cochrane Risk of
 445 Bias Tool is not designed for use with non-randomized controlled trial, post-trial study,
 446 retrospective study, single-arm study or case report. Finally, eleven of the twenty
 447 studies were RCTs, which were conducted under strict surveillance. Such studies may
 448 not accurately reflect real-world vision care, therefore more real-world studies are
 449 needed to better understand long-term safety of RLRL in a greater variety of settings.

450

451 **Conclusion**

452 In conclusion, no irreversible visual function loss or ocular structural damage
 453 associated with RLRL therapy was identified in this review. Meticulous supervision is
 454 crucial throughout the entire treatment process by clinicians and parents, including
 455 documenting the status of retina through fundus photography and OCT before
 456 initiating RLRL therapy and at each routine examination, as well as tracking visual
 457 acuity and the duration of any afterimages at home. Screening for and early
 458 identification of rare super-responders is also important to avoid potential light injury.
 459 Future larger and longer-term real-world studies are needed to better understand the

460 long-term safety of RLRL.

461

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465

466 **Author Contributions:**

467 Prof. Wang had full access to all the data in the study and took responsibility for the
468 integrity of the data and the accuracy of the data analysis.

469 Study concept and design: Chen, Congdon, Wang.

470 Data collection: Chen, Chen, Xiong, Yang, Wang.

471 Analysis and interpretation of data: Chen, Wang.

472 Drafting of the manuscript: Chen, Congdon, Wang.

473 Critical revision of the manuscript for important intellectual content: All authors.

474 Statistical analysis: Chen, Wang.

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476 Administrative, technical, or material support: Congdon, Wang.

477 Overall supervision: Wang.

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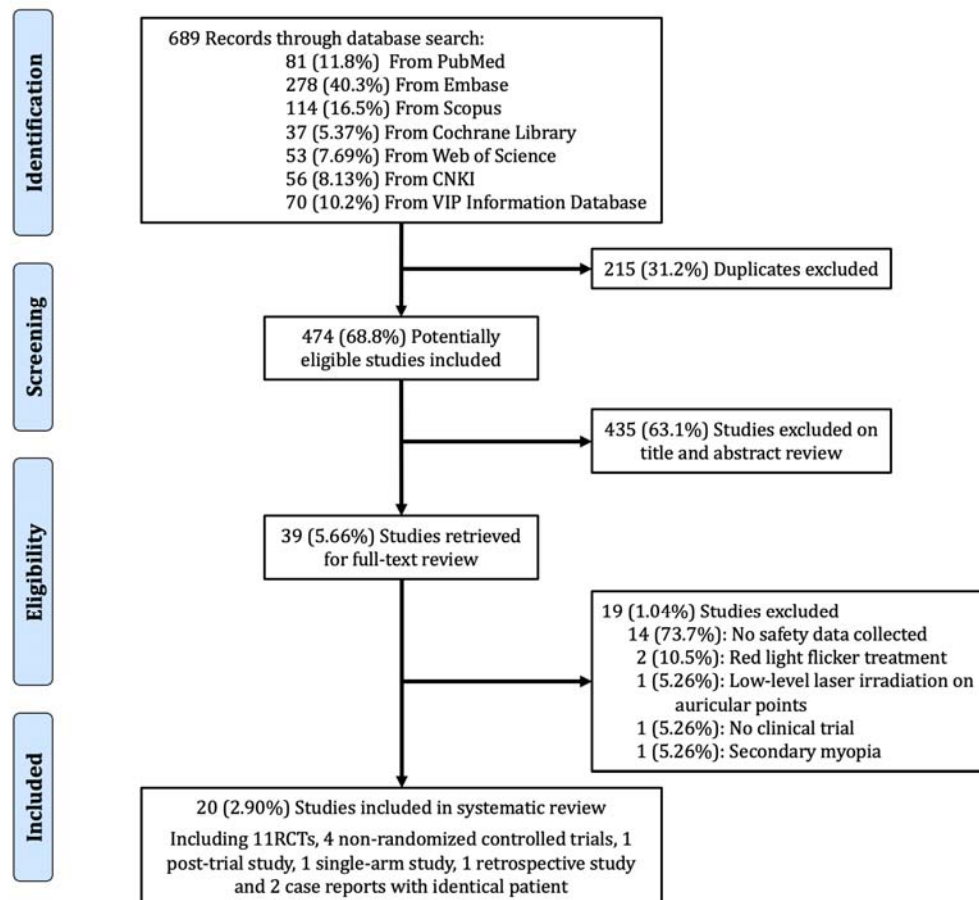
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Figure Legends

Figure 1. Study selection outlined according to Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines. RCT=randomized controlled trial.



761 **Figure 2.** Risk of bias assessment by Cochrane Risk of Bias Tool of the included
762 RCTs. RCT=randomized controlled trial.

	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel (performance bias)	Blinding of outcome assessment (detection bias)	Incomplete outcome data (attrition bias)	Selective reporting (reporting bias)	Other bias
Chen (a) et al 2022	+	+	-	+	+	+	+
Chen (b) et al 2022	+	?	-	+	+	+	?
Dong et al 2022	+	+	+	+	+	+	+
He et al 2023	+	+	-	+	+	+	+
Jiang et al 2021	+	+	-	+	+	+	+
Liu (a) et al 2024	+	?	-	+	+	?	?
Tian (a) et al 2023	+	?	-	+	+	?	+
Tian et al 2022	+	?	-	+	+	?	+
Yan et al 2021	+	?	-	+	+	+	+
Zhou (a) et al 2023	+	+	-	+	+	?	+
Zhou (b) et al 2023	+	?	-	+	+	?	+

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Table 1. Characteristics of included studies.

No. of Patients by Study (Study Type)	Treatment	Age (y)	Female (%)	Mean SER (D)	Mean AL (mm)	Completion Rate (%)
Jiang et al 2021 (RCT) ⁸						
117	Intervention (RLRL)	8-13	62 (52.1)	-2.49 (0.92)	24.5 (0.67)	98.3
129	Control (SVS)	8-13	72 (49.7)	-2.67 (1.06)	24.6 (0.86)	89.0
Yan et al 2021 (RCT) ³³						
60	Intervention (RLRL)	7-12	58 (48.3)	-2.52 (1.15)	24.2 (0.92)	100
60	Control (SVS)	7-12		-2.53 (1.15)	24.4 (0.79)	100
Xiong et al 2022 (post-trial follow-up study) ⁹						
11	RLRL-RLRL	8-13	4 (36.4)	-1.77 (0.57)	24.9 (0.94)	-
10	SVS-RLRL	8-13	5 (50.0)	-2.57 (1.11)	24.8 (0.71)	-
52	RLRL-SVS	8-13	26 (50.0)	-2.50 (0.83)	24.5 (0.58)	-
41	SVS-SVS	8-13	26 (63.4)	-2.76 (1.15)	24.6 (0.94)	-
Dong et al 2022 (RCT) ¹¹						
56	Intervention (RLRL)	7-12	30 (53.6)	-3.13 (1.91)	24.7 (1.04)	94.6
55	Control (Sham device)	7-12	26 (46.4)	-2.82 (1.86)	24.6 (0.96)	98.2
Chen (a) et al 2022 (RCT) ³⁴						
30	Intervention (RLRL)	7-15	17 (54.8)	-2.60 (1.17)	24.5 (0.79)	96.8
30	Control (0.01% atropine)	7-15	14 (45.2)	-2.59 (1.24)	24.7 (0.98)	96.8

Chen (b) et al 2022 (RCT) ³⁵						
46	Intervention (RLRL)	6-13	19 (41.3)	-2.54 (1.04)	24.6 (0.97)	90.2
40	Control (SVS)	6-13	15 (37.5)	-2.29 (0.77)	24.6 (0.76)	78.4
Tian et al 2022 (RCT) ³⁶						
91	Intervention (RLRL)	6-12	55 (49.1)	-2.00 (0.33)	24.3 (0.92)	81.3
88	Control (SVS)	6-12	57 (50.9)	-2.00 (0.25)	24.2 (0.85)	78.6
He et al 2023 (RCT) ¹⁰						
126	Intervention (RLRL)	6-11	68 (48.9)	0.14 (0.30)	23.4 (0.68)	90.7
122	Control (SVS)	6-11	71 (51.1)	0.16 (0.28)	23.3 (0.69)	87.8
Zhou (a) et al 2023 (RCT) ³⁷						
20	Intervention (RLRL)	3-16	9 (37.5)	-2.93 (1.87)	24.6 (1.16)	96.0
15	Control (SVS)	3-16	7 (41.2)	-2.11 (1.21)	24.4 (0.87)	76.0
Zhou (b) et al 2023 (RCT) ³⁸						
43	Intervention (RLRL 0.37mW)	6-15	20 (40.0)	-1.72 (0.91)	24.2 (0.79)	86.0
47	Intervention (RLRL 0.60mW)	6-15	25 (50.0)	-2.01 (0.87)	24.1 (0.88)	94.0
44	Intervention (RLRL 1.20mW)	6-15	24 (48.0)	-2.08 (1.33)	24.5 (0.91)	88.0
43	Control (SVS)	6-15	21 (42.0)	-2.10 (0.90)	24.4 (0.90)	86.0
Tian (a) et al 2023 (RCT) ³⁹						
56	Intervention (RLRL)	6-12	33 (58.9)	0.25 (0.25)	23.1 (0.80)	100
56	Control (SVS)	6-12	31 (55.4)	0.25 (0.19)	23.1 (0.70)	100
Lin et al 2023 (non-randomized controlled trial) ⁴¹						

41	Intervention (RLRL)	6-18	84 (51.2)	-3.20 (2.82)	24.3 (1.04)	58.6
58	Intervention (RLRL)	6-18		-7.93 (2.95)	25.7 (1.57)	82.9
65	Control (SVS)	6-18		-2.32 (2.64)	24.3 (1.21)	92.9
Zhao et al 2023 (non-randomized controlled trial) ⁴²						
47	Intervention (RLRL)	6-18	18 (38.3)	-2.31 (1.26)	24.6 (0.88)	100
20	Control (SVS)	6-18	8 (40.0)	-2.75 (0.84)	24.8 (0.90)	100
Wang et al 2023 (retrospective study) ⁴⁶						
434	Intervention (RLRL)	3-17	200 (46.1)	-3.74 (2.60)	24.9 (1.20)	-
Liu et al 2023 & Tian (b) et al 2023 (case report) ^{47,48}						
1	Intervention (RLRL)	12	1 (100)	OD -6.75D OS -6.25D	-	-
Liu (a) et al 2024 (RCT) ⁴⁰						
47	Intervention (RLRL)	7-12	19 (44.2)	0.17 (0.35)	23.6 (0.78)	91.5
47	Control (SVS)	7-12	20 (47.6)	0.30 (0.35)	23.3 (0.73)	89.4
Liu (b) et al 2024 (single-arm study) ⁴⁵						
40	Intervention (RLRL)	7-14	17 (42.5)	-2.75 (1.43)	24.9 (0.97)	100
Xiong et al 2024 (non-randomized controlled trial) ⁴³						
45	Intervention (RLRL)	7-16	22 (48.9)	-3.00 (0.90)	23.6 (0.35)	100
45	Control (SVS)	7-16	23 (51.1)	-3.02 (0.11)	23.6 (0.37)	100
Zhang et al 2024 (non-randomized controlled trial)* ⁴⁴						
44	Intervention (RLRL)	≥7	25 (56.8)	-1.87 (1.16)	24.1 (0.80)	100

32	Intervention (OK)	≥ 7	14 (43.8)	-2.44 (1.15)	24.2 (0.73)	100
29	Intervention (RLRL+OK)	≥ 7	10 (34.5)	-2.55 (1.32)	24.5 (0.96)	100
36	Control (SVS)	≥ 7	18 (50.0)	-2.11 (1.11)	24.3 (0.82)	100

* Only eye number for each group and total participant number was available in the publication.

Data were presented as number (percentage) or mean (standard deviation).

RLRL=repeated low-level red-light; SVS=single vision spectacles; RCT=randomized controlled trial; SER= spherical equivalent refraction;

AL=axial length; OK=orthokeratology.

Table 2. Visual function, ocular structures and adverse events reported in RLRL studies.

Study	Follow-up (mo)	Visual function outcome	Ocular Structural Outcome	Adverse events
Jiang et al 2021 ⁸	12	108 (97.3%) participants achieved BCVA of 20/20, and the BCVA in the remaining 3 (2.7%) participants was 20/25. *	No structural damage on photosensory layer was identified by OCT. (n=72)	• No adverse events were reported, including glare, flash blindness, or afterimages. • No severe adverse events developed, including sudden vision loss by 2 lines occurring in a period of a few second/s or minutes to a few days or scotoma.
Yan et al 2021 ³³	12	-	-	• No ocular adverse events were reported, including conjunctival hyperemia, edema, photophobia, tears, corneal epithelial damage, lens opacity, macular damage. • No cognitive impairment and behavioral abnormalities were found.
Xiong et al 2022 ⁹	12	10 (100%) participants maintained BCVA of 20/20.	No structural damage on photosensory layer was identified by OCT.	• No severe adverse events or side effects including sudden vision loss of more than two lines, scotoma, dazzling, short-term glare, flash blindness, and afterimages developed.
Xiong et al 2022 ⁹	24	11 (100%) participants maintained	No structural damage on	No severe adverse events or side effects

		BCVA of 20/20.	photosensory layer was identified by OCT.	including sudden vision loss of more than two lines, scotoma, dazzling, short-term glare, flash blindness, and afterimages developed.
Dong et al 2022 ¹¹	6	-	-	18 adverse events were reported in 13 participants, but none were related to RLRL therapy. †
Chen (a) et al 2022 ³⁴	12	58 eyes (100%) maintained BCVA of 20/20.	No structural damage was identified by OCT.	• No severe adverse events were reported, including blindness, death, hospitalization, or conditions requiring medical or surgical interventions. • No adverse events including a sudden vision loss of two lines or more, a scotoma, photophobia, allergy, dry mouth, or tachycardia developed. • 1 participant reported dizziness after the red-light therapy, but the symptoms resolved after a few minutes and only occurred in the immediate post-treatment period for a few days.
Chen (b) et al 2022 ³⁵	12	BCVA remained normal at each follow-up visit.	No structural damage including vitreomacular traction, macular schisis,	No adverse events including dazzling, glare, long-term afterimages, and flash blindness were reported.

			macular hole, intraretinal fluid, subretinal fluid, hemorrhage, retinal pigment epithelium proliferation, and atrophy was identified by OCT.	
Tian et al 2022 ³⁶	6	-	-	No adverse events including photophobia, eye itching, burning sensation, dry eye, blurred vision, glare, dazzling, keratitis, and conjunctivitis were reported.
He et al 2023 ¹⁰	12	123 (100%) participants achieved BCVA of 20/25.	No structural damage including vitreomacular traction, macular schisis, macular hole, intraretinal fluid, subretinal fluid, hemorrhage, retinal pigment epithelium proliferation, and atrophy was identified by OCT.	No adverse event was reported, including glare, flash blindness, or afterimages longer than 6 minutes after treatment. ††
Zhou (a) et al 2023 ³⁷	12	No BCVA indicated visual function loss.	No structural damage was identified by OCT.	• Reversible subjective symptoms without objective ocular abnormalities (CTCAE grade 1) ‡, including reversible vision

				loss lasting 2.1 ± 0.7 min (n=23) after red-light therapy due to flash blindness or glare with afterimage, and afterimage with an average of 3.2 ± 1.2 min. • No dry eye, cataract, keratitis, night blindness, photophobia or any other permanent visual impairment. • No systemic adverse effects (such as headache or dizziness), severe adverse events, or other adverse events related to grades of ocular diseases: Grade 2–5 of CTCAE.
Zhou (b) et al 2023 ³⁸	6	BCVA remained normal at each follow-up visit.	No retinal or choroidal structural abnormalities were identified by OCT.	No adverse events were reported.
Tian (a) et al 2023 ³⁹	6	-	-	No adverse events were reported.
Lin et al 2023 ⁴¹	2	-	-	• Afterimage is the most common phenomenon post RLRL therapy, which can be alleviated by a short period of eye-closing and rest; with the progression of treatment, this phenomenon does not appear to worsen.

				· No adverse events were reported.
Zhao et al 2023 ⁴²	1	-	No retinal structure related photodamage was identified by OCT.	-
Wang et al 2023 ⁴⁶	At least 12	-	-	No adverse events were reported.
Liu et al & Tian (b) et al 2023 ^{47,48}	5	• Binocular BCVA declined from 20/20 to 20/30. § • Multifocal electroretinogram revealed the response of the central macula (ring 1) was moderately decreased and the response of paramacula (ring 2 and 3) was mildly decreased in both eyes.	• Bilateral foveal ellipsoid zone disruption and interdigitation zone discontinuity were identified by OCT. § • Fundus photographs revealed bilaterally darkened foveae with a hypoautofluorescent plaque in autofluorescence images.	Prolonged afterimages after light therapy (occasionally exceeding 8 minutes).
Liu (a) et al 2024 ⁴⁰	12	No BCVA indicated visual function loss.	No structural damage was identified by OCT.	No adverse events were reported.
Liu (b) et al 2024 ⁴⁵	6	-	No change in macular structure or microcirculatory system was identified by OCT and OCTA.	No adverse events were reported.
Xiong et al 2024 ⁴³	6	-	No ocular surface damage	No adverse events were reported.

			was found, and no structural damage or macular hemorrhage was identified by OCT.	
Zhang et al 2024 ⁴⁴	6	BCVA remained unchanged at each follow-up visit compared to baseline.	No structural damage within 6mm macular region was identified by OCT.	No adverse events were reported.

RLRL=repeated low-level red-light therapy; mo=month; BCVA=best corrected visual acuity; OCT=optical coherence tomography.

*In the SVS control arm, the proportion of compromised BCVA 20/25 was 8 of 112 (7.1%).

†18 adverse events included one case each of influenza, an eyelid injury requiring surgical repair, and acute mesenteric lymphadenitis; two cases each of earwax blockage, and allergic rhinitis; and three cases of dental caries.

††Two participants dropped out due to the afterimage duration exceeded 6 minutes at baseline.

‡Common Terminology Criteria for Adverse Events (CTCAE) grade 1 refers to mild; asymptomatic or mild symptoms; clinical or diagnostic observations only; intervention not indicated.

§After 3 months without RLRL therapy, the bilateral outer retinal damage partially recovered, and the visual acuity improved to 20/25 in both eyes. After 4 months discontinued RLRL therapy, the bilateral outer retinal damage totally recovered.

Table 3. Estimated NNH and NNH/NNT ratios associated with RLRL therapy.

Variable	Multiplier	Value*
Annual incidence of vision loss		8.77
Years of vision loss accrued	× 65 years	570
NNH†		
For 1 year of vision loss	10,000/years vision loss	17.5
For 5 years of vision loss	5 × 10,000/years vision loss	87.7

* We assumed that the annual incidence of vision loss from RLRL therapy as 8.77 per 10,000 patient-years. Any vision loss is estimated to be experienced for 65 years after the event.

† The NNH/NNT ratios are 12.7-fold for -3D, 14.4-fold for -4D, 16.4-fold for -5D, 18.1-fold for -6D, 19.9-fold for -7D and 21.4-fold for -8D with RLRL therapy.

All values are presented per 10,000 patients.

RLRL=repeated low-level red-light; NNH=number needed to harm; NNT=number needed to treat.

Table 4. Comparison of the ocular adverse event incidence rate between RLRL and other interventions.

Interventions	Baseline age criteria (years)	Number of participants	Person- years	Ocular adverse events	Ocular adverse events per 100 patient-years	95% CI*
RLRL ^{8-11,33-48}	3-18	1436	1139.8	1	0.088	0.02-0.50
Low-dose atropine ⁶⁰⁻⁷⁸	4-16	2736	5368	393	7.32	6.65-8.05
Orthokeratology ^{76,79-84}	6-12	231	364.5	75	20.6	16.7-25.0
Anti-myopia contact lens ⁸⁵⁻⁹⁰	7-15	697	1899	366	19.3	17.6-21.1
Spectacles designed for myopia reduction ⁴⁹⁻⁵⁹	6-16	1247	2299	5	0.22	0.09-0.51

RLRL=repeated low-level red-light; CI=confidence interval.

Anti-myopia contact lens includes rigid gas permeable contact lens and soft multifocal contact lens. Spectacles designed for myopia reduction includes bifocal lens, progressive addition spectacles, aspherical lenslets and peripheral defocus spectacles.

*95% CI was calculated using Wilson methods.